

To Aid Emotion Expression for Non-Verbal Autistic Individuals, In an App, Less (Features) is More (Better)

Jamey Albert
Western Washington University
Bellingham, WA, USA

Daniel Koronhály
Western Washington University
Bellingham, WA, USA

Samantha Dobesh
Western Washington University
Bellingham, WA, USA

Shameem Ahmed
Western Washington University
Bellingham, WA, USA

Moushumi Sharmin
Western Washington University
Bellingham, WA, USA

Zafir Nasim
Allen High School
Allen, TX, USA

Ilaana Khan
Westlake Highschool
Austin, TX, USA

ABSTRACT

A significant proportion of autistic children are considered ‘non-verbal or minimally verbal’ and have difficulty expressing their emotions. Commercial alternative and assistive communication (AAC) apps are often inaccessible and ineffective in aiding autistic children in expressing emotions. Lack of access to adequate AACs is more prevalent in lower and middle-income communities (LMICs), which neither have sufficient infrastructure to aid the development of autistic children nor the resources to make expensive technology affordable. We propose the design of a prototype app that is accessible and multi-modal and caters to the specific needs of non-verbal and minimally verbal autistic children. Our design is grounded in research in autism and strength-based technology design. We incorporate accessibility guidelines and universal design features to make our design appropriate and effective for target users.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; *Ubiquitous and mobile computing*.

KEYWORDS

Autism Spectrum Disorder (ASD); alternative communication; emotions; expression

ACM Reference Format:

Jamey Albert, Daniel Koronhály, Samantha Dobesh, Shameem Ahmed, Moushumi Sharmin, Zafir Nasim, and Ilaana Khan. 2022. To Aid Emotion Expression for Non-Verbal Autistic Individuals, In an App, Less (Features) is More (Better). In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3544793.3560338>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9423-9/22/09.

<https://doi.org/10.1145/3544793.3560338>

1 INTRODUCTION

Research highlights that autistic children experience difficulty in self-expression. While most children acquire language during their preschool years, a significant minority remains ‘non-verbal or minimally verbal.’ An estimated 40 percent of autistic children fail to develop fully functional verbal means of communication [1], which can lead to poorer social skills [6], challenging behavior [18] such as social withdrawal [8], and overall poorer quality of life. About half of autistic individuals also experience alexithymia, or the inability to comprehend one’s emotions, often leading to anxiety, depression, and other communication problems [7] [13]. Table 1 shows prevalence of this condition. Table 1 highlights that a significantly large proportion of the autistic population resides in the lower and middle-income communities (LMICs), who face this challenge to a greater extent. Though AACs exist, autistic children and their families in LMICs are commonly unable to use them due to language constraints (AAC applications predominantly use English), high cost, and expensive hardware required to use the apps. We aim to address these challenges by designing an affordable mobile app to aid non-verbal and minimally verbal autistic children that will cater to their specific needs by using a strength-based design.

Table 1: ASD and Non-Verbal Based on Location

		Nonverbal ASD	LMIC
Diagnosed ASD (world)	with	≈ 30% [6]	≈ 95% [12]
Diagnosed ASD (USA)	with	≈ 28% [15]	

2 CURRENT STATE OF ART AND DESIGN METHODOLOGY

To understand the landscape of available technology for supporting emotion expression for non-verbal and minimally verbal autistic children, we analyzed applications that are available commercially on Google and Apple app stores. Our research reveals that communication support apps deviating from the cookie-cutter vocabulary grids are scarce, and commercial apps often lack needed accessibility features. While expressing emotions is a critical part of social

communication, apps designed to aid emotion expression are extremely rare. To bridge this gap, we identified design features for an accessible and effective mobile app that will aid autistic children in expressing their emotions. We incorporated accessibility guidelines pertaining to autistic children as their needs and challenges vary significantly from those of neurotypical children.

2.1 Design Methodology

First, we identified three major mediums that can potentially aid non-verbal and minimally verbal autistic children in emotion expression: music (non-speech sounds), color, and symbols. Our choice is inspired by the strength-based approach [11], which advocates focusing on target users' strengths rather than weaknesses. This approach is supported by research on technology design for specific user groups. This selection of three mediums also enhances usability, as autistic children have varied strengths and preferred modes of communication. We analyzed these three components of emotion expression by conducting systematic literature reviews on individual components. We found relevant papers using queries in databases such as PubMed, ACM, and Google Scholar. For example, to identify papers on the topic of emotion and music, we used the following query: ("music therapy" OR "musical therapy" OR "music intervention") AND ("self expression"). This specific query generated 62 papers. Similar queries were used to identify relevant research about color and symbols as mediums of emotional expression. We analyzed the papers to generate design guidelines for usability and accessibility. We conducted an informal survey which asks to identify emotions co-related with 16 emojis, aiming to understand their generalizability and effectiveness. 22 participants who varied in ethnicity (Asian, Hispanic, and White), age (9 to 82 years, majority in their 20s), and gender (male, female, non-binary) participated in the survey.

3 DESIGNING TO AID EMOTION EXPRESSION

3.1 Music and Emotion Expression

Research on music and self-expression seeks to evaluate the emotional connections that children make with different music qualities. Many musical parameters are directly associated with the intensity of emotions. Table 2 lists music parameters and their relationships with different emotional states. Children are able to distinguish, from an early age, emotions communicated through music. At the age of three, children can distinguish happy and sad emotions in music; by age six, most children can identify happy, sad, anger, and fear in music [16]. As many non-verbal autistic children use non-speech sounds to express themselves and often show an affinity for music, we incorporated a set of musical tunes for users to express their emotions [19].

3.2 Color and Emotion

Colors used in apps are a critical component in their universality, accessibility, and usability as research shows that certain colors are associated with specific emotions [21]. Children relate emotions to color, connecting bright colors to positive feelings and darker colors to negative feelings [5]. Additionally, colors are associated with emotions and emotion-laden words [20], indicating the importance of using colors with their corresponding emotions to

Table 2: Music Findings Table [16]

Parameter	Happy	Sad	Angry
tempo	fast	slow	fast
volume	med-high	low	high
pitch	high	low	high
articulation	staccato	legato	staccato
timbre	bright	dull	sharp
noise	none	none	high
modality	major	minor	minor

utilize and affirm existing associations, as a contradiction would be confusing and frustrating for users. The principle of consistency suggests that users awareness of these associations aids learnability. Though at present there is no standardized color-to-emotion mapping, there are specific color-emotion mappings that are favored by researchers and designers of assistive technology for autism. For example, JeStiMulE used the color-emotion association used in Plutchik's Wheel of Emotions to relate the six basic emotions to colors [17]. Other AAC apps use different but similar color-emotion mappings. These findings inspired us to use an appropriate color-to-emotion mapping. The colors chosen for the seven emotions are as follows: happiness = yellow, sadness = blue, anger = red, neutral = gray, disgust = green, fear = black, surprise = orange, and in pain = brown [20] [17]. In addition, it is not uncommon for autistic individuals to experience color hypersensitivity. For example, bright colors may cause eye strain due to their high luminance [9]. To increase usability and customization, three different color palettes are advised: two of these use soft, muted colors with minimal vibrancy, chosen to alleviate potential eye-strain.

Table 3: A subset of the recommended guidelines [4]

W3C Category	Guideline	Applied?
Focus and Structure, Consistency	The general design and the structure should be simple, clear and predictable, secondary content that distracts the user should be avoided, and number of features available at any time should be limited	✓
Multi-Modality	Pictures can be drawings, photographs, symbolic images, should be easy to understand, should not go in the background, should be in a sharp focus.	✓
Readability and Language	The text should go with pictures. It should be clear, simple, and short; should be in a big font (14), in plain Sans-serif style (e.g., Verdana), and in a mild color.	✓

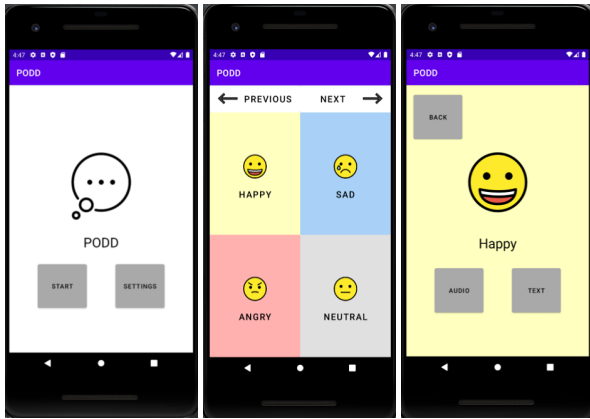


Figure 1: The home page of the application; a "feelings page," with four feeling buttons; the singular "Happy" feeling page, with Audio (music) and Text (text-to-speech) buttons.

3.3 Symbolic Representation of Emotion

The survey results confirmed our assumption that emojis accurately convey their intended emotions. Of the seven core emotions chosen (happiness, sadness, anger, neutral, disgust, fear, surprise, and pain), most were correctly identified by participants by the emojis. Some of the more complex emotions, such as surprise and fear, had lower agreement percentages on the correct word, indicating that such symbols need better visual representation to convey the specific emotions. Our results support findings that state that emoticons are sufficient and effective in communicating emotions [14] [10].

3.4 Guidelines to Design Applications for Autistic Individuals

There are specific guidelines that indicate design choices that optimize the usability and accessibility of apps targeted to autistic users [2] [3]. We analyzed and condensed these guidelines; a subset of these is presented in Table 3 along with whether specific guidelines were followed in our design [4]. For example, W3C guidelines recommend simple, clear, and predictable interfaces where available features should be limited and clutter-free, consistent use of interactions, and when possible, using copious pictures and multiple representations to support information retention and visibility. Table 3 shows screens in our app that use large, universal representations of information, refrain from presenting too much information in any screen, and use consistent interaction through out the app.

4 CONCLUSION AND FUTURE DIRECTION

Our proposed app was designed to aid non-verbal or minimally verbal autistic children in emotion expression. We utilized colors, music, and symbols for emotional communication, to not only optimize accessibility and usability but also give users an option to choose their preferred mode of communication. We designed the app to be affordable and accessible for individuals living in the LMICs, using mobile devices as the target platform. Future work includes refining our app by conducting usability evaluations and testing to seek feedback from our target users. We also aim to provide an intuitive way for users to alter the music parameters

listed in Table 2 while providing a set of default musical notes that serve as an accessible way for inexperienced users to express emotion.

REFERENCES

- [1] 2021. Autism statistics and facts. <https://www.autismspeaks.org/autism-statistics-asd> Accessed: 07-14-2022.
- [2] 2021. Cognitive Accessibility User Research. <http://w3c.github.io/coga/user-research/#autism>. Accessed: 07-14-2022.
- [3] Yuska Paola Costa Aguiar, Edith Galy, Anaïs Godde, Maëla Trémaud, and Carole Tardif. 2022. AutismGuide: a usability guidelines to design software solutions for users with autism spectrum disorder. *Behav. Inf. Technol.* 41, 6 (April 2022), 1132–1150.
- [4] Jamey Albert. 2022. ASD Application Guidelines Table. <https://docs.google.com/spreadsheets/d/1gvR5V6hqnFhHmP8jOeGy-kFaFaOf1TMGCvhhMeQDVM/edit?usp=sharing>
- [5] C J Boyatzis and R Varghese. 1994. Children's emotional associations with colors. *J. Genet. Psychol.* 155, 1 (March 1994), 77–85.
- [6] Amanda Brignell, Karen V Chenausky, Huan Song, Jianwei Zhu, Chen Suo, and Angela T Morgan. 2018. Communication interventions for autism spectrum disorder in minimally verbal children. *Cochrane Database Syst. Rev.* 11, 11 (Nov. 2018), CD012324.
- [7] Laura Dattaro. 2020. Difficulty identifying emotions linked to poor mental health in autistic people. *Spectrum* (2020).
- [8] Nancy C. Brady John F. Connolly Nicole Gage Barry Gordon Portia Iversen Connie Kasari Rebecca Landa Janice Light Catherine Lord E. Mark Mahone Stewart H. Mostofsky Rhea Paul MaryAnn Ronski Laura Schreibman Laura Schreibman Helen Tager-Flusberg Grace T. Baranek, April A. Benasich. 2010. Who remains non-verbal? What are the characteristics of this population? *National Institute on Deafness and Other Communication Disorders*. (April 2010).
- [9] Marine Grandgeorge and Nobuo Masataka. 2016. Atypical color preference in children with autism spectrum disorder. *Front. Psychol.* 7 (Dec. 2016), 1976.
- [10] Kun-An Hsiao and Pei-Ling Hsieh. 2014. Age Difference in Recognition of Emotions. In *Lecture Notes in Computer Science*. Springer International Publishing, 394–403. https://doi.org/10.1007/978-3-319-07863-2_38
- [11] Matthew Jones, Marita Falkmer, Ben Milbourn, Tele Tan, Sven Bölte, and Sonya Girdler. 2021. Identifying the essential components of strength-based technology clubs for adolescents with autism spectrum disorder. *Dev. Neurorehabil.* 24, 5 (July 2021), 323–336.
- [12] Aubrey J Kumm, Marisa Viljoen, and Petrus J de Vries. 2022. The digital divide in technologies for autism: Feasibility considerations for low- and middle-income countries. *J. Autism Dev. Disord.* 52, 5 (May 2022), 2300–2313.
- [13] Bethany F M Oakley, Emily J H Jones, Daisy Crawley, Tony Charman, Jan Buitelaar, Julian Tillmann, Declan G Murphy, Eva Loth, and EU-AIMS LEAP Group. 2022. Alexithymia in autism: cross-sectional and longitudinal associations with social-communication difficulties, anxiety and depression symptoms. *Psychol. Med.* 52, 8 (June 2022), 1458–1470.
- [14] Anna Oleszkiewicz, Tomasz Frackowiak, Agnieszka Sorokowska, and Piotr Sorokowski. 2017. Children can accurately recognize facial emotions from emoticons. *Computers in Human Behavior* 76 (2017), 372–377. <https://doi.org/10.1016/j.chb.2017.07.040>
- [15] Annio Posar and Paola Visconti. 2021. Update about "minimally verbal" children with autism spectrum disorder. *Rev. Paul. Pediatr.* 40 (Sept. 2021), e2020158.
- [16] Suvi Saarikallio, Mari Tervaniemi, Antti Yrtti, and Minna Huottilainen. 2019. Expression of emotion through musical parameters in 3- and 5-year-olds. *Music Education Research* 21, 5 (Oct. 2019), 596–605. <https://doi.org/10.1080/14613808.2019.1670150>
- [17] Sylvie Serret, Stephanie Hun, Galina Iakimova, Jose Lozada, Margarita Anas-tassova, Andreia Santos, Stephanie Vesperi, and Florence Askenazy. 2014. Facing the challenge of teaching emotions to individuals with low- and high-functioning autism using a new Serious game: a pilot study. *Mol. Autism* 5, 1 (July 2014), 37.
- [18] Jeff Sigafos. 2000. *Education and Training in Mental Retardation and Developmental Disabilities* 35, 2 (2000), 168–176.
- [19] Michael J. Silverman. 2008. Nonverbal Communication, Music Therapy, and Autism: A Review of Literature and Case Example. *Journal of Creativity in Mental Health* 3, 1 (May 2008), 3–19. <https://doi.org/10.1080/15401380801995068>
- [20] Tina M Sutton and Jeanette Altarriba. 2016. Color associations to emotion and emotion-laden words: A collection of norms for stimulus construction and selection. *Behav. Res. Methods* 48, 2 (June 2016), 686–728.
- [21] Patricia Valdez and Albert Mehrabian. 1994. Effects of Color on Emotions. *Journal of Experimental Psychology: General* 123, 4 (1994), 394–409. <https://doi.org/10.1037/0096-3445.123.4.394>